Name of the module: mixer
Target delivery date:
Location in GIT repository: vspa_common\vspa-lib\mixer

Overview:

This module is called in two situations:

1. Shift 40 or 80MHz WiFi signal in frequency domain and then extract primary 20-MHz signal for LSTF detection;
2. Mix input signal with a frequency offset given by "carrier frequency offset estimate" module to let RX has the same frequency grid as TX.

Function API:

```
extern unsigned int mixer_asm(
   vspa2_complex_fixed16 *py,
   vspa2_complex_fixed16 *px,
   uint32_t PhaseIn,
   int FreqIn,
   size_t L
);
```

API description:

Mixer operation is implemented via elementwise vector multiplication in time domain: **y** = **x**.***m**, where **x** is the input signal, **m** is an NCO-generated complex sequence, and **y** is the mixed signal.

Function inputs:

px	Pointer to x, the signal to be mixed. Vector-aligned. In complex 16-bit half-fixed.
PhaseIn	initial phase to NCO, which used to generate m. Unsigned 32-bit int.
FreqIn	base frequency to NCO, which used to generate **m.** 1's comp signed 32-bit int.
L	The number of input DMEM lines. L >= 1.

Function outputs:

py	Pointer to y, the mixed output. Vector-aligned. In complex 16-bit half-fixed. The function supports in-place if py=px.
Return value	The NCO phase ready for mixing the next batch of input samples. Unsigned 32-bit int.

DMEM requirements:

Variable	Minimum size (words)	Minimum alignment (words)	Allocated by
y	32	32 (vector-aligned on 16AU)	Caller

x	32	32	Caller
L	1	1	Caller

Implementation:

Implementation is ported from mixer in Rayleigh 4AU library. Theoretically, one line of cHF input requires {1 load, 1 store} and {2 cmad}, hence, 2 cycle per input line.

The following is to explain how NCO uses "PhaseIn" and "FreqIn" to generate a complex sequence. Mixer application configures NCO in "normal" mode. In 16AU, NCO configured in normal mode generates a 16-point complex exponential sequence **m** by:

$$\boldsymbol{m} = e^{\frac{j2\pi kf(i+16n+[0,1,2,\ldots,15])}{2^{32}}},$$

where

- *k* is base frequency multiplier in uint16,
- *f* is base frequency in int32 with 1's complement,
- *i* is initial phase in uint32,
- n is the batch index. Every read of S1 increases n by 1.

The implementation configures NCO as such:

Parameters of NCO	**Value**
k	1
f	FreqIn
i	PhaseIn
n	starts with 0 when first entering the mixer function

PhaseOut is 16*2L when the function is called to mix L lines such that the next call to mixer can pick up the history phase and continue the mixing.

Test plan:

L = {1,2,3,4,5,6,7,8}.
FreqIn = [-32 -16 16 64] ppm.
PhaseIn = 0 and 2^32 - [170 1 253 17 16 2] to test Phase wrap-up. When phase is wrapped to 0, NCO will not create a glitch because $\frac{j2\pi kf(2^{32})}{2^{32}}$ is always a multiple of 2π.

References:

VSPA_16AU_ISA_v2.0_ISM, Section 4.20 "Numerically controlled oscillator (NCO) instructions"